Maize (corn) - Gray leaf spot

RAG record ADV-002 | canonical key: maize_gray_leaf_spot

Record ID	ADV-002
Crop / crop key	Maize (corn) / maize
Condition / disease key	Gray leaf spot / gray_leaf_spot
Main pathogen	Cercospora zeae-maydis / Cercospora zeina
RAG aliases	maize gray leaf spot, maize grey leaf spot, Cercospora zeae-maydis, Cercospora zeina, GLS

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Early spots are small tan or brown flecks. Later they become long, narrow, rectangular grey-to-tan lesions limited by the leaf veins.
- When disease is severe, many lesions join together and large leaf areas dry up.

Weather or field conditions that favour it

Warm humid weather, heavy dew and infected maize residue left on the soil increase risk.

What to do now

- Start with field sanitation, crop rotation and resistant hybrids because the fungus survives in maize residue.
- If fungicide is economically justified, treat early. Research shows the best response when spraying begins while disease is still low and mainly on lower leaves.
- Because a current India-specific label dose for maize gray leaf spot was not verified in the official sources used here, the app should not invent a dose.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Propiconazole /	Use the exact rate on a	These active ingredients	[S1,S2]

Chemical / formulation	Rate from cited source	How to use this information	Source
difenoconazole / azoxystrobin / tebuconazole	currently registered Indian maize/GLS product label	have shown efficacy against maize gray leaf spot in published research. The RAG system should present them as research-supported options, not as automatic CIBRC approval.	

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Rotate away from maize where disease is repeatedly severe.
- Manage infected residue and avoid continuous maize in high-risk fields.
- Choose resistant hybrids where available.
- Scout lower leaves before tasselling and during humid weather.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] FAO AGRIS / Plant Disease literature record. Frequency and timing of fungicide applications for control of gray leaf spot in maize. https://agris.fao.org/search/en/providers/122535/records/65ded93a4c5aef494fde4f62

[S2] Embrapa Maize & Sorghum. Fungicide efficacy against Cercospora zeae-maydis in maize. https://www.embrapa.br/en/busca-de-publicacoes/-/publicacao/487789/avaliacao-da-eficiencia-de-fungicidas-no-controle-da-cercosporiose-cercospora-zeae-maydis-na-cultura-do-milho